



Self-powered co-harvesting for batteryless sensing and monitoring in two-wheeled IoT applications



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G. Pasquini, M. Doglioni, A. Saviane and D. Brunelli (2025), "Self-Powered Co-Harvesting for Batteryless Sensing and Monitoring in Two-Wheeled IoT Applications," in IEEE STAR 2025.

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Cycling is becoming an IoT platform

Electronics are now part of modern bikes

Sensors and
telemetry

Wireless controls
and actuators



This integration delivers tangible improvements across critical functional domains:

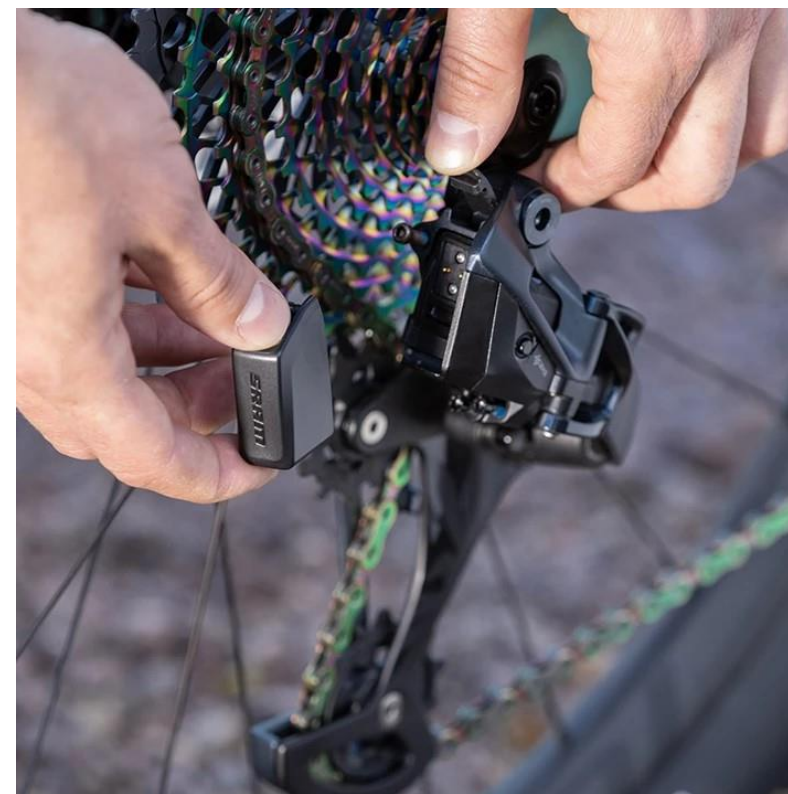
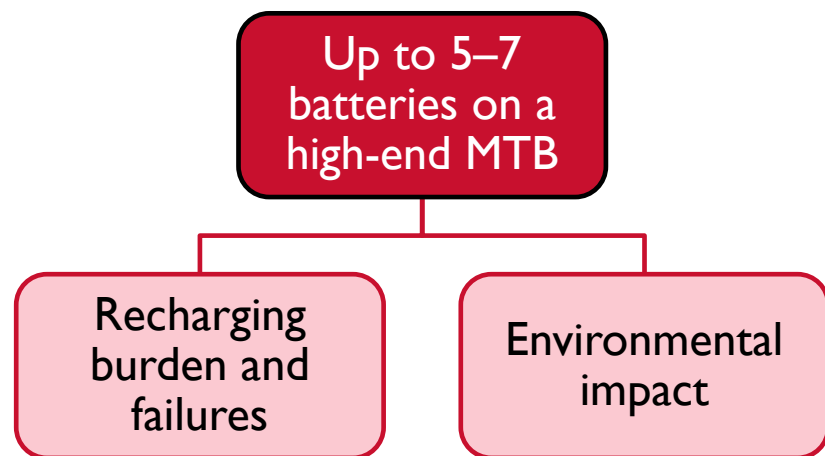
- Real-time **performance monitoring**
- Improved **safety enhancement**
- Smarter **bike setup optimization**



BYB telemetry & Fox Transfer Neo Remote controller



The problem: too many batteries



We need autonomy, not more batteries

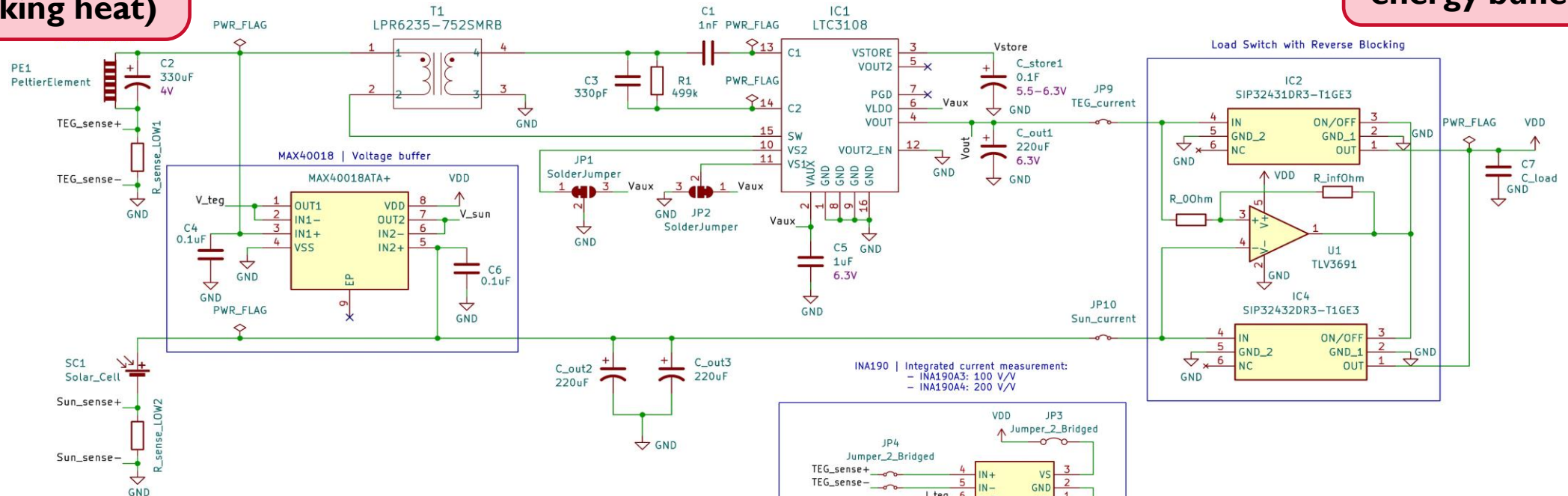
Core idea

- Hybrid harvesting using **TEG** on brake pads + small **PV cells**
- The two sources are **naturally complementary** in mountain biking
- Provides **friction-free energy** under highly variable conditions
- Harvesting adapts to both braking events (TEG) and ambient light (PV)
- **Power sources also act as sensors** (TEG $\leftrightarrow$ brake temperature $\leftrightarrow$ braking events)

Hybrid system concept

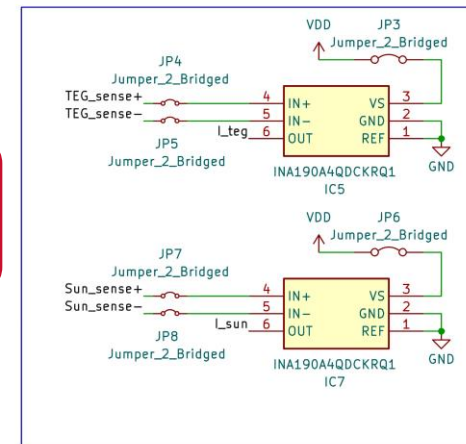
TEG
downhill bursts
(braking heat)

Shared
energy buffer



PV
continuous
background
energy

Power
measurement



Solar harvesting (PV)

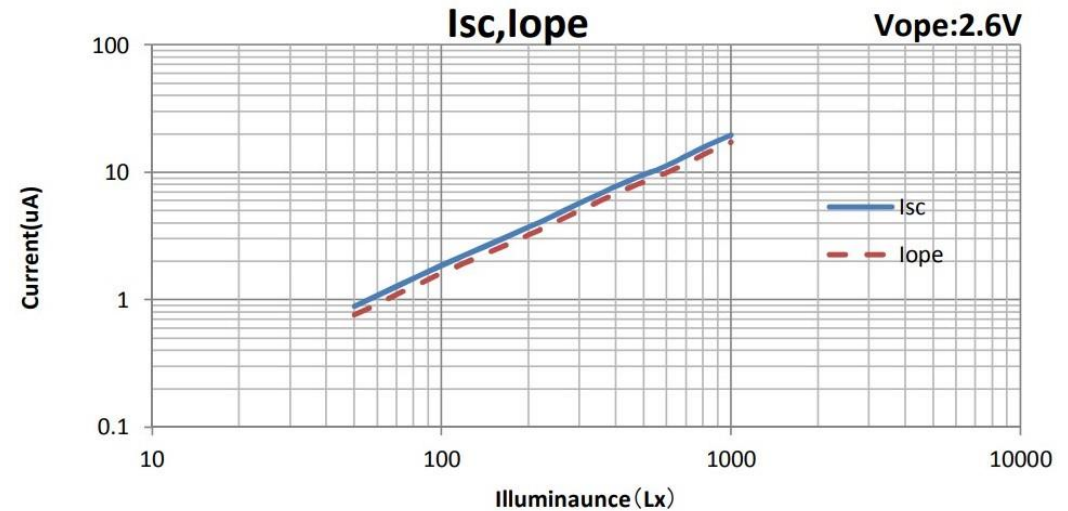
- Specific **PV** for **low-light conditions**, including cloudy or indoor environments
- Generates **μW** to **mW** depending on sunlight
- Provides **baseline power** over the full ride



PV (AM-1606C-MEL), mounted on BFTAG01. A sensor tag based on STM32L0

Output power across common lighting conditions:

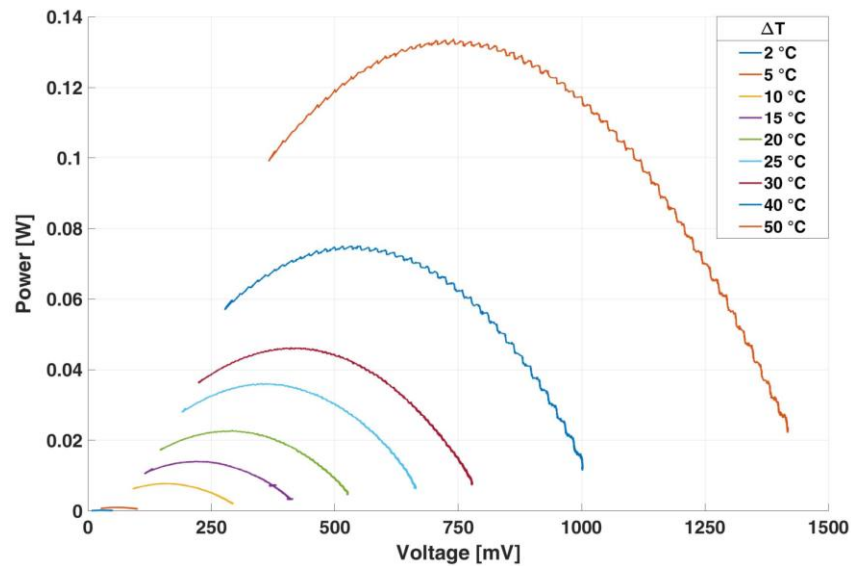
- **Indoor/shade:** $\sim 8\text{--}30\ \mu\text{W}$
- **Daylight:** $\sim 170\ \mu\text{W}$
- **Direct sunlight:** $\sim 1\ \text{mW}$



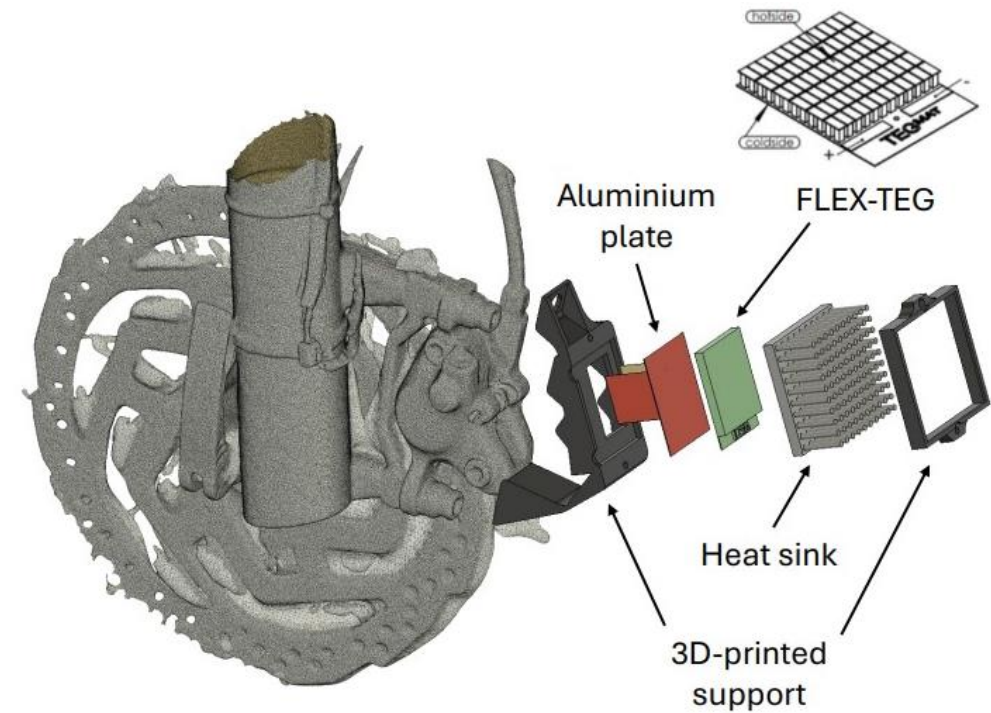
PV Current vs. Illuminance at Vo

Thermoelectric harvesting (TEG)

- Mounted on the brake caliper using the natural **temperature gradient (ΔT)**
- Generates **mW-level bursts** during braking events
- Output directly follows **thermal dynamics**



Lab characterization, power generation depending on ΔT .



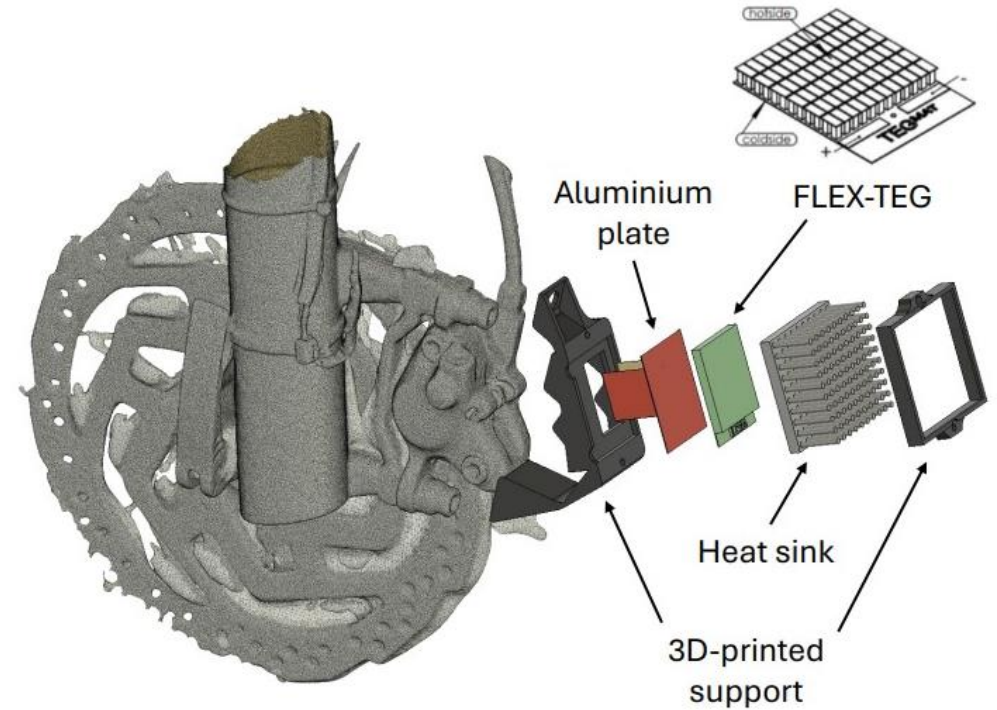
TEG supporting structure

Real TEG setup on the bike

- Custom **3D-printed mount**
- **Aluminum plate** in direct contact with the brake caliper (hot side)
- **Heat sink on the cold** side to maximize ΔT and harvesting efficiency
- TEG mounted **without friction** on the wheel system



TEG setup

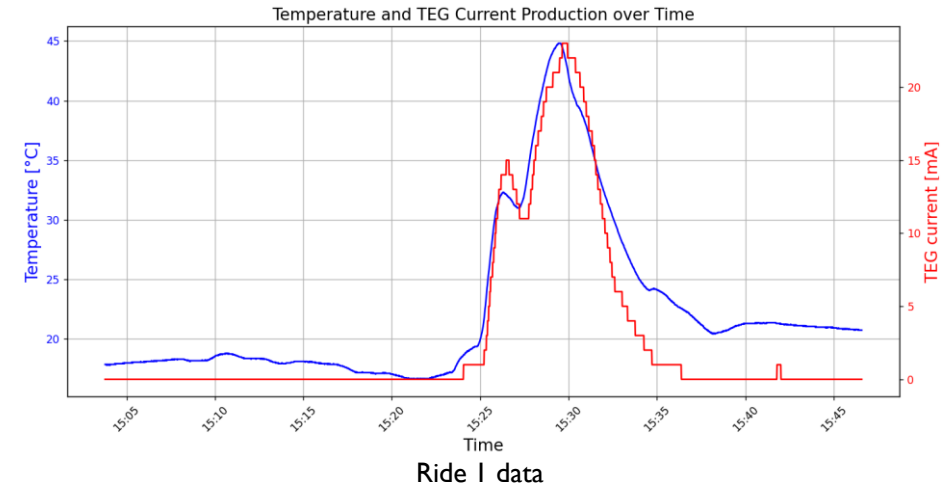


TEG supporting structure

TEG on-field results

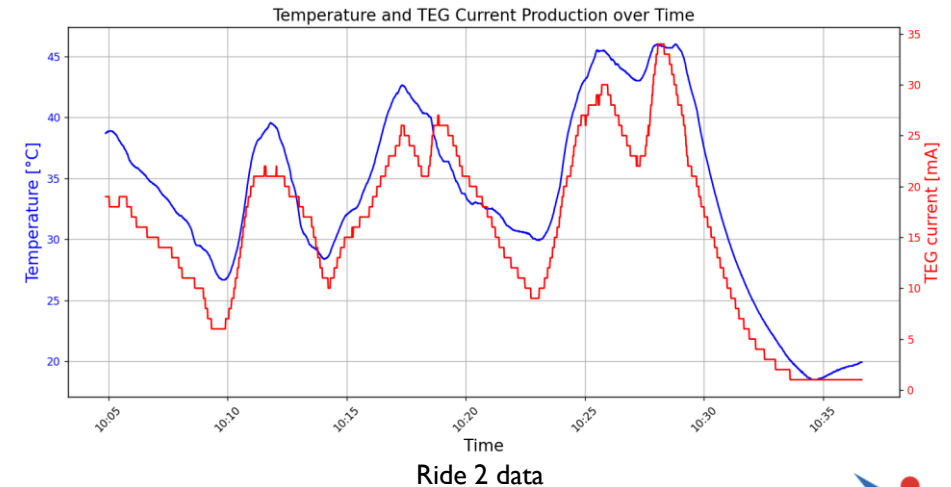
Ride 1

- 1.5 km downhill segment
- Maximum TEG hot-side temperature: 45°C
- Ambient temperature: ~20°C
- **Peak power: 2.3 mW**
- **Harvested energy: 439 J**



Ride 2

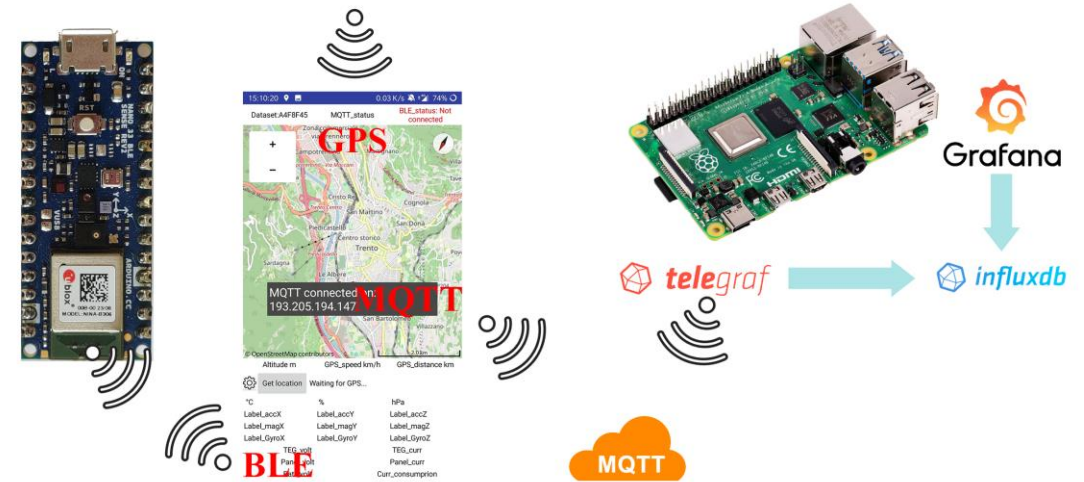
- longer descent (~6 km),
- Maximum TEG hot-side temperature: 46°C
- Ambient temperature: ~16°C
- **Peak power: 3.2 mW**
- **Harvested energy: 2236 J**



What we do with the energy: Two Use Cases

Use Case #1: IoT pipeline for batteryless telemetry

- **Bike setup & configuration**
- **Performance** monitoring
- Track **real-time data** on usage
- **Data visualization & insights**



BLE → Smartphone App → Cloud (MQTT) → Dashboard for Real-time data and visualization

Use Case #2: Power commercial actuators and remote controllers

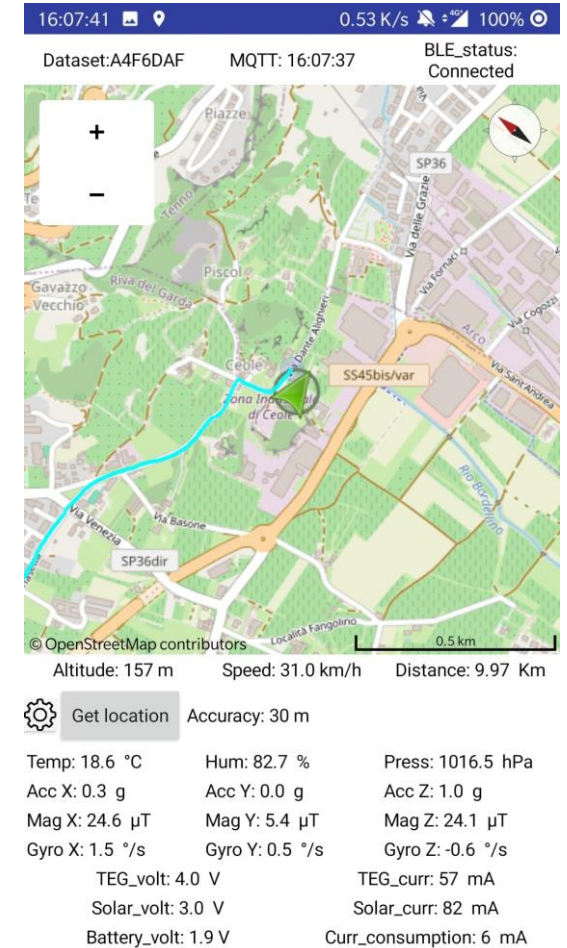
- Electronic shifting systems
- Seatpost droppers

Use Case #I: Batteryless Telemetry

Harvested energy powers a self-sustaining sensor node for sensing, processing and wireless transmission.

Strategy: Energy is buffered on a capacitor and the node wakes only when sufficiently charged

- **Required energy per cycle: 0.133 mJ** Brake temperature, IMU, suspension
- Continuous sampling during downhill (TEG with $\Delta T > 20\text{ }^{\circ}\text{C}$)
- 0.77 s sampling period with PV only (10.000 lux)



Use Case #2: Powering Commercial Devices

- **Goal:** Use harvested energy to drive commercial on-board actuators and wireless components (today battery-powered)
- **Works today** (low-power): remotes, shifters, power meters (**1–4 mW**)
- **Not yet** (high-power): droppers, derailleurs, active suspension (**30–300 mW**)
- Device optimization is key: a dropper can work battery-free with short, limited pushes powered by a capacitor, but the trade-off is reduced continuous actuation.
- Continuous actuation: needs optimization.



Key takeaways

Hybrid harvesting

stable & reliable power
profile

Results

Zero friction, zero
maintenance, zero charging

Telemetry

batteryless today (fully
supported)

Commercial devices

partially supported
(optimization needed)

Conclusions & future work

- **Improve heat transfer and TEG efficiency** to approach lab-scale performance in real riding conditions
- Add more energy sources like **vibration energy harvesting** for rough terrains and MTB use cases
- Extend the architecture to e-bikes and **additional sensors** for telemetry
- Move toward fully batteryless on-bike electronics for maintenance-free IoT systems